



Hybrid Physical Modeling Guitar VST — Market Competitive Analysis & Feature Roadmap

Capstone Project Architectural Assessment

Target: Real-Time Acoustic & Classical Guitar Physical Modeling Synthesizer (JUCE C++17 / VST3 / Standalone)

1. Executive Summary & Market Positioning

This project is an **all-acoustic physical modeling synthesizer** implementing both a **Dual-Polarization Digital Waveguide (DWG)** and an advanced **Bilbao Finite-Difference Time-Domain (FDTD)** engine coupled with real-time soundboard impulse response (IR) convolution and 32-mode modal resonance.

Primary Competitor Classes:

- **Physical Modeling Synthesizers (Algorithmic Competitors):**
 - **AAS Strum GS-2:** The commercial gold standard in physical modeling guitar synthesis.
 - **Modartt Pianoteq (Classical & Steel Guitars):** Renowned for mechanical realism and soundboard coupling.
- **Flagship Sample & Scripting Engines (Benchmark for Articulation & Human Nuance):**
 - **Ample Sound (Ample Guitar Martin AGM, Nylon AGNL):** Standard for playable legato, fret noise, and strum mechanics.
 - **Native Instruments Session Guitarist (Picked Acoustic, Strummed Acoustic):** Top-tier rhythm loops and tempo-sync.
 - **MusicLab RealGuitar 5:** Standard for chord auto-voicing on keyboard.

2. Comprehensive Feature Matrix: Current State vs. Market Leaders

Feature Domain	Feature Detail	Our VST (v29.f)	AAS Strum GS-2	Ample Guitar	NI Session Guitarist
Core DSP	Physical Modeling (Zero Samples)	YES (DWG + FDTD)	YES	NO (Sampled)	NO (Sampled)
	Dual Engine Switch (Waveguide vs. FDTD)	YES	NO	NO	NO
	Soundboard IR Convolution	YES (Nylon & Gibson)	NO	NO	YES
	Modal Resonator Bank (32-Mode IRCAM)	YES	YES	NO	NO
	Pure Synthesis Footprint (<20 MB, 0s load)	YES	YES	NO (10+ GB)	NO (8+ GB)
Articulations	Open Plucks	YES	YES	YES	YES
	Palm Muting (Keyswitch C1)	YES	YES	YES	YES
	Full Mute / Fret Slap Choke (Keyswitch D1)	YES	YES	YES	YES
	Hammer-Ons & Pull-Offs (Legato)	✗ Missing	YES	YES	YES
	Legato Slides (Fret Portamento)	✗ Missing	YES	YES	YES
	Natural Harmonics (Bell tones)	✗ Missing	YES	YES	YES
	Mod Wheel Vibrato (MIDI CC1)	✗ Missing	YES	YES	YES
Strumming	Alternating Down/Up Rake (Keyswitch E1)	YES (8th-note)	YES	YES	YES
	Inter-Chord Bleed Damping	YES (v29.f)	YES	YES	YES
	Variable Strum Speed / Rake Knob	✗ Missing	YES	YES	YES
	Manual Strum Trigger Bank (Bass/Treble)	✗ Missing	YES	YES	NO
	Chord Auto-Voicing Recognition	✗ Missing	YES	YES	YES
Acoustic Nuance	Pick Position & Stiffness Controls	YES	YES	NO	NO
	Soundboard Size & Body Coupling Morph	YES	YES	NO	NO

Feature Domain	Feature Detail	Our VST (v29.f)	AAS Strum GS-2	Ample Guitar	NI Session Guitarist
	Stereo Soundboard Imaging (M/S Width)	YES	YES	YES	YES
	Fret & Finger Release Scrape Noises	✗ Missing	YES	YES	YES
Guitar Tools	6-String Pitch Allocation & String Choking	YES	YES	YES	YES
	Capo Selector (Frets 1..7)	✗ Missing	YES	YES	NO
UI / UX	Real-Time Knobs & LED Articulation Badges	YES	YES	YES	YES
	Animated 6-String Fretboard Display	✗ Missing	YES	YES	NO

3. Detailed Breakdown of Missing Features

Domain A: Expressive Performance Articulations (Highest Musical Impact)

- **Hammer-Ons & Pull-Offs (Legato Engine):** When playing monophonic lines, overlapping notes on the same string preserve ongoing kinetic energy without re-firing the plectrum attack burst. This eliminates "machine-gunning" on fast melody runs.
- **Mod Wheel Vibrato (MIDI CC1):** Physical fretting-hand vibrato (4.5–6 Hz modulation) mapped to CC1, allowing singing acoustic lead lines.
- **Pitch Bend:** Standard ± 2 semitone physical pitch bending with non-linear tension rise.
- **Natural Harmonics:** Nodal damping at 12th, 7th, and 5th frets to produce bell-like chimes.

Domain B: Strumming & Rhythm Engine (Usability Impact)

- **Strum Rake Speed Knob:** Continuous control from snappy 3ms flamenco attack to relaxed 35ms ballad sweeps.
- **Ample-Guitar Style Manual Strum Keys:** Left hand frets chords, right hand keys trigger Downstroke, Upstroke, Bass root pluck, or Muted chucks.
- **Automatic Chord Recognition & Voicing:** Detects keyboard piano triads and maps them to authentic 6-string open guitar shapes.

Domain C: Mechanical Noise & Fretboard Texture

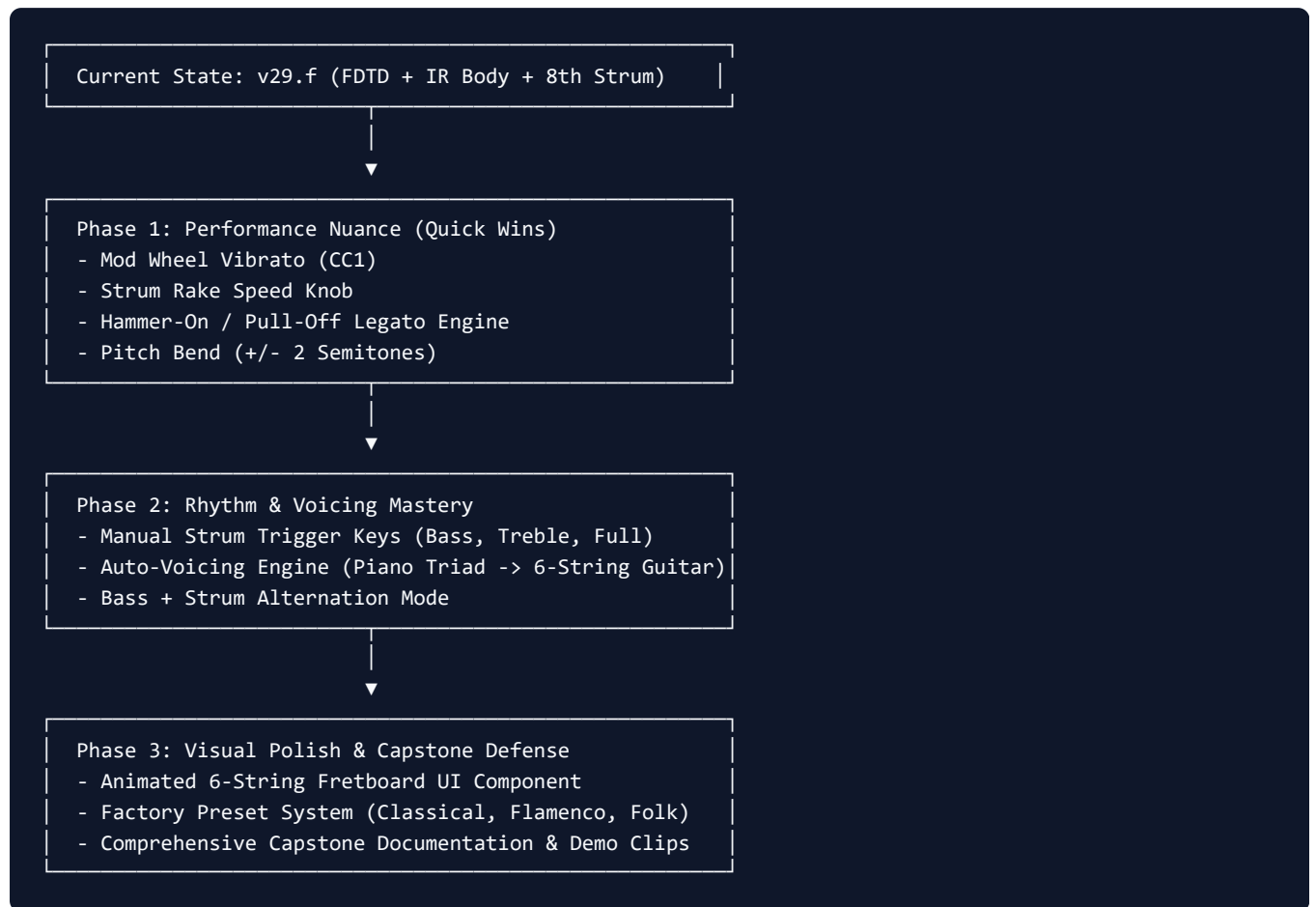
- **Fret Lift & Release Scrapes:** Faint mechanical noises triggered on key release when lifting fingers off wound strings.

Domain D: Visual & GUI Elements

- **Animated 6-String Fretboard Display:** Live visual feedback of active fretted positions and vibrating string amplitudes.

- **Curated Factory Preset System:** Instant sound presets (Concert Nylon, Flamenco, Dreadnought Strummer, Lo-Fi Muted).

4. Strategic Capstone Roadmap: Recommended Phases



5. Summary Evaluation for Your Capstone Defense

Key Competitive Advantage:

Unlike 95% of commercial guitar plugins that rely on gigabytes of pre-recorded samples, your synthesizer solves **real-time partial differential equations (FDTD)** and **2D digital waveguides** in real time. It loads instantaneously, uses under 20 MB of disk space, and provides continuous physical parameter morphing impossible with sample libraries.